

Ali Haydar Sucu

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Education

Ankara Yıldırım Beyazıt University

B.Sc. in Computer Engineering

Ankara, Türkiye

Sep 2023 - Jun 2028 (Expected)

- Current GPA: 3.12/4.00
- Activities: AYBÜ Hezarfen Student Club, Developer Students Community, IEEE Student Branch

Professional Experience

T3 Gemstone (Baykar • T3 Foundation)

Embedded Software Intern

Istanbul, Türkiye

Jul - Aug 2026

- Developed and integrated ROS 2 Humble software for the Bumin unmanned ground vehicle on the T3 Gemstone O1 board, including LiDAR, RF2O odometry, SLAM Toolbox, IMU, motor encoder, GPIO and PWM-based motor-control components.
- Built and tested an autonomous indoor room-mapping pipeline in Gazebo/Rviz and Docker, adapting frontier-driven exploration to the existing navigation stack for later real-hardware deployment.
- Participated in pre-packaging QA of nearly 500 Gemstone O1 boards in a five-person team; used TTL, multimeter, oscilloscope and continuity testing to isolate USB, connector and board-level faults.
- Contributed to hardware/software integration, technical documentation and the ChibiOS + ArduPilot porting roadmap for the Cortex-R5 real-time core.

AYBÜ ÖMER

AI Developer / Part-time Student

Ankara, Türkiye

May 2025 - Jun 2026

- Led development of the BuraiTech - Ayaz Project, an intelligent document-interactive AI agent using the Llama 4 LLM and a web interface deployed for university-wide access.
- Assigned development tasks, conducted prompt engineering and designed a modular architecture; optimized inference with caching strategies and embedding reuse.
- Designed the system architecture for real-time audio streaming and document embedding, enabling retrieval-grounded question answering, summarization and paraphrasing.
- Deployed the assistant on a public-facing kiosk at AYBÜ ÖMER's Milli rade Campus location, where students can query programs and services through a student-facing chat interface.

Ankara Yıldırım Beyazıt University

Bootcamp Tutor & Student Mentor

Ankara, Türkiye

Aug 2024 - May 2025

- Led a summer bootcamp introducing university students to front-end web development with HTML and CSS, supporting participants as they built functional websites.
- Provided technical support and mentorship to engineering freshmen through the "Meet Your Student Tech Staff" program.
- Represented the university at international conferences and supported the Learning and Teaching Center's technology-focused activities.

Engineering Projects

Autonomous Unmanned Ground Vehicle

Terranova UGV - Embedded Software Developer

Ankara, Türkiye

2026 - Present

- Created two custom Gazebo simulation worlds, establishing repeatable environments for autonomous rover testing.
- Developed a browser-based ground-control station with live vehicle markers, sensor cards and telemetry/control interfaces; tested the integration with a standard Gazebo rover.
- Implemented ROS 2 nodes for IMU, LiDAR, ultrasonic and GPS sensors, and contributed to the system architecture, technical reports and integration planning for the six-wheel rocker-bogie, skid-steering platform.

Autonomous Unmanned Sea Vehicle

Göksu USV - PUSULA Team, Embedded Software Developer

Ankara, Türkiye

Oct 2024 - Aug 2025

- Developed autonomous navigation and control software on NVIDIA Jetson using Python and ROS 2, integrating LiDAR, ZED 2 stereo camera, Pixhawk and GPS data streams.
- Engineered systemd services and real-time node integration; collaborated with the electronics team on hardware/software co-design, field testing and sensor communication debugging.
- Reached the finals and placed 10th among 284 teams in Teknofest's Mavi Vatan competition with the PUSULA USV team.

Rocket Avionics and Embedded Systems

Gökçe Rocket Project - AYBÜ HEZARFEN Club, Avionics Team Member

Ankara, Türkiye

Sep 2023 - Jun 2024

- Researched and evaluated suitable sensors for the rocket avionics system.
- Contributed to flight-control board development and assisted with sensor and component integration.
- Performed initial avionics-system testing with the team and documented the system for Teknofest.

Selected Software Projects

Autonomous Room Mapping with a Mobile Robot

2026

Bumin UGV - T3 Gemstone Internship Project | ROS 2, Nav2, Gazebo, Python/C++

GitHub: github.com/alihaydarsucu/t3gemstone-bumin-ika

- Built a modular autonomous exploration stack that detects frontier regions, ranks candidate goals by utility and tracks frontier state across exploration cycles.
- Integrated the exploration loop with Nav2 planning, SLAM Toolbox and a browser-based GCS for live monitoring and teleoperation.
- Maintained a shared launch and deployment workflow across Dockerised Gazebo simulation and T3 Gemstone O1 hardware, enabling simulation-to-hardware validation.

Local Retrieval-Augmented AI Assistant

2026

Burai - document AI, FAISS, local LLM inference, Python

GitHub: github.com/alihaydarsucu/burai-tech-ayaz

- Developed a privacy-oriented assistant for institutional documents with PDF, DOCX and TXT processing, FAISS-based retrieval and local Ollama inference.
- Implemented retrieval-grounded question answering, summarization and paraphrasing through a modular pipeline designed for local deployment.

STM32F103RB Embedded Systems Practice

2024 - 2025

STM32CubeIDE, C, HAL, ARM Cortex-M3, FreeRTOS

GitHub: github.com/alihaydarsucu/stm32f103-practices

- Implemented and tested firmware on the NUCLEO-F103RB covering GPIO, UART, PWM, ADC, EXTI, I2C and FreeRTOS using STM32 HAL.
- Practiced peripheral configuration, timer-based control, interrupt-driven firmware and RTOS task design through independently testable examples.
- Used STM32CubeIDE and ST-LINK-based debugging to build, flash and validate embedded applications on the ARM Cortex-M3 platform.

Personal Portfolio and Automated CV System

2024 - Present

LaTeX, GitHub Actions, HTML/CSS, CI/CD

GitHub: github.com/alihaydarsucu/alihaydarsucu.github.io

- Built a portfolio with automated CV generation and GitHub project synchronization workflows, demonstrating reproducible document builds and deployment automation.

Focus and Activity Tracking for Linux

2025

FocusPath - C++, Qt, X11, SVG, Linux

GitHub: github.com/alihaydarsucu/FocusPath

- Built a low-level Linux desktop application that captures X11 window events and logs activity as JSONL for focus and distraction analysis.
- Generated SVG timelines, pie charts and usage reports from recorded desktop activity.

Batch GitHub Repository Puller

2025

GitHubPuller - Python, GTK4, libadwaita, Linux

GitHub: github.com/alihaydarsucu/GitHubPuller

- Developed a native Linux desktop utility for batch-cloning and updating multiple GitHub repositories from one interface.
- Built the desktop UI with GTK4 and libadwaita, focusing on clear repository selection and repeatable update workflows.

Technical Skills

Programming: C/C++, Python, Bash/Shell, JavaScript, Java

Robotics: ROS 2, Nav2, SLAM Toolbox, Gazebo, Docker, sensor integration, autonomous navigation

Embedded: STM32, ARM architecture, GPIO/UART/PWM, RTOS concepts, ChibiOS, NuttX, Jetson

Tools: Linux, Git, GitHub Actions, STM32CubeIDE, ST-LINK, FastAPI

Languages: Turkish (Native), English (Professional)

Certifications & Achievements

Teknofest Success Certificate

Nov 2025

Teknofest

Ranked 10th among 284 teams in the Mavi Vatan competition

Career Module Certificate

Jul 2025

MSMB - Mimar Sinan Mühendisler Birlii

Professional career module

Data Science Bootcamp Certification

Dec 2023

Google Developer Students Community

Python, NumPy, Pandas and Matplotlib

Volunteering

MMCE Mühendisler Dernei

Secretary & Member, University Council

Ankara, Türkiye
Sep 2025 - Present

- Led Secretariat operations, managing formal communication with academic units and university council representatives.
- Planned and coordinated engineering-focused events and student development activities with the council team.
- Established structured record-keeping and follow-up processes to preserve continuity across council decisions.
- mimce.org

Research & Presentations

Language as a Tool: Challenge-Based Summer Bootcamp Model

Co-author and Presenter

International / Türkiye
2024 - 2025

- “Language as a Tool: A Challenge-Based Summer Bootcamp Model with Scrum Scaffolding to Develop Global Competencies” - poster presented at the International Challenge-Based Learning Conference, Eindhoven University of Technology, Netherlands, Apr 2025.
- “Bootcamps: Language as a Tool to Develop Global Competencies” - paper presented at the 6th FLE Joint Conference Unity in Diversity, Ankara, Türkiye, May 2025.
- “Language as a Tool: A Summer Bootcamp Model to Build Global Competencies” - paper presented at the 6th International English for Specific Purposes Conference, METU, Türkiye, Nov 2024.
- “Fostering Autonomy through CLIL-Infused Bootcamps and Scrum Scaffolding” - pre-recorded presentation at the 13th Lab Session on CLIL and Learner Autonomy, Kanda University of International Studies, Japan, Nov 2024.
- “Sharpen Your Arguments: Fallacy Fighters Workshop” - paper presented at the 5th FLE Joint Conference Unity in Diversity, AYBÜ, Ankara, Türkiye, Jun 2024.
- “Beyond Self: E-Portfolios and Ecological L2 Motivation” - paper presented at the 5th FLE Joint Conference Unity in Diversity, AYBÜ, Ankara, Türkiye, Jun 2024.